# SHQ 3D

3D DESIGN & 3D PRINTING SOLUTIONS

## MECHATRONICS & 3D PRINTING PORTFOLIO

*by: @ariqshaq*

SHQ 3D

# SERVICE OFFER

## > 3D DESIGN

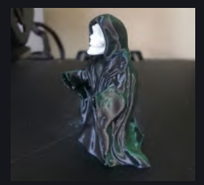
- We provide **3D Design by Request**, specializing in **Technical Design and Product Design** for your projects.
- Experienced in **Mechatronics Design**, where we provide a requested **Detailed Blueprint or Technical Drawing**, including component placement and selection.
- All designs are **Supported for Autodesk Softwares**.

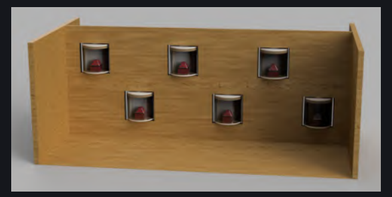
## > 3D PRINTING

- We provide **FDM 3D Printing** supported for **PLA, PETG, and TPU Materials**.
- Your submitted designs can be **Consulted First or Adjusted by Us for Free**, so your designs fit well with the printing tolerance.
- **Guaranteed to Print Immediately** after the designs are submitted.
- We open to **Requests for Printing Settings**.

## TRACK RECORD

- **Delivering Solutions Across Various Projects**, including lab research, campus projects, company requests, and competition clients.
- **Experienced with Over 3+ Years**, handling tight deadlines and complex specification requests.
- **Multiple Nationwide Recognitions** from national competitions to high-stakes projects.





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Contacts:

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SELECTED PROJECTS

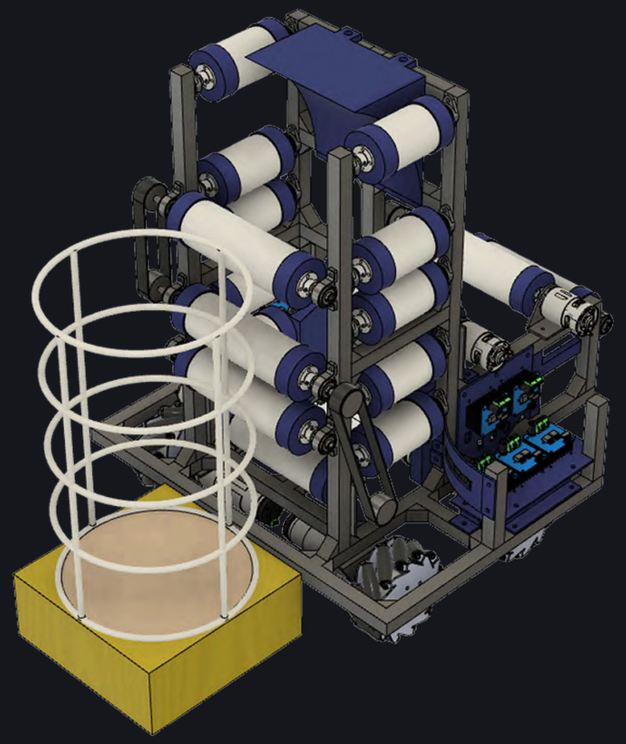
# ABU ROBOCON 2024 ROBOT

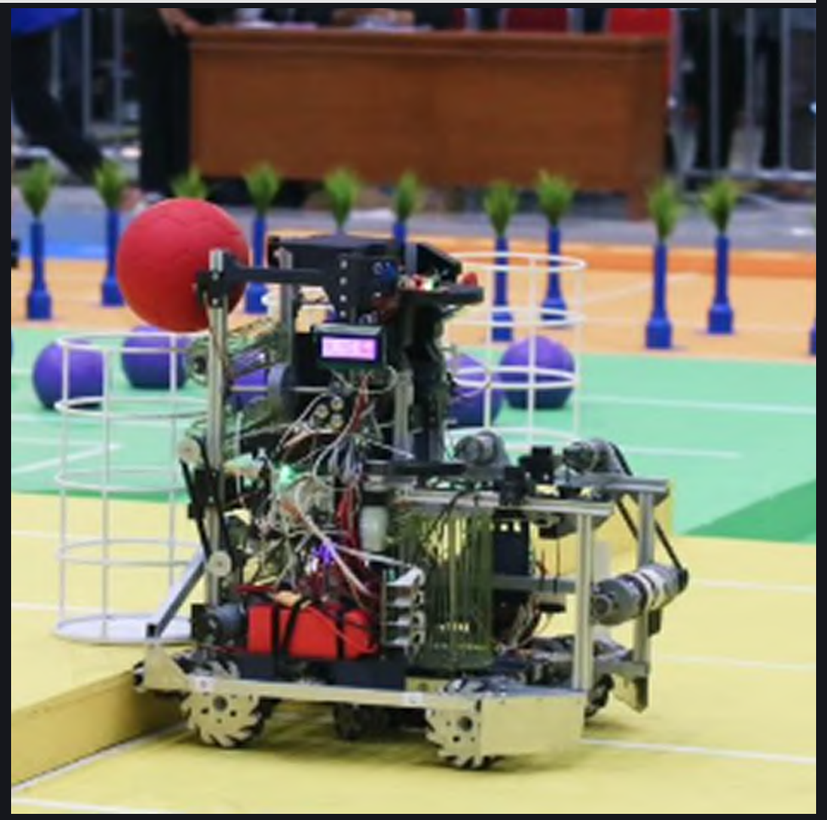
## DESCRIPTION:

**An Automatic Ball Sorting Robot**, capable of picking and storing selected colored balls into silos.

## TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Grinding
- Drilling
- Hand Tools





## ROLE:

Mechanics (3D Design, Component Selection, Assembly)

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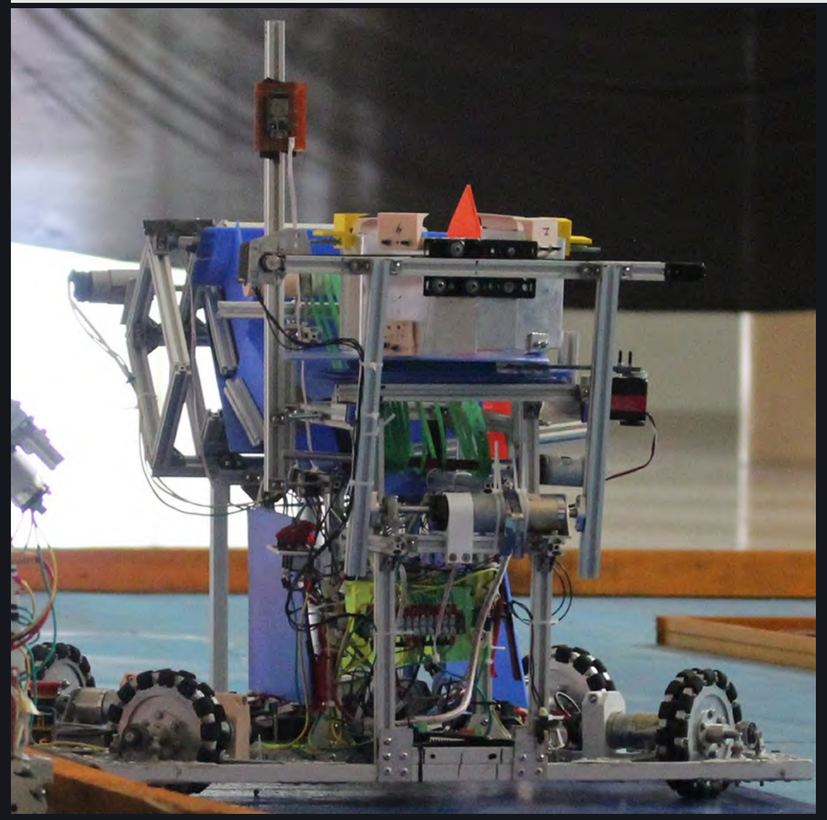
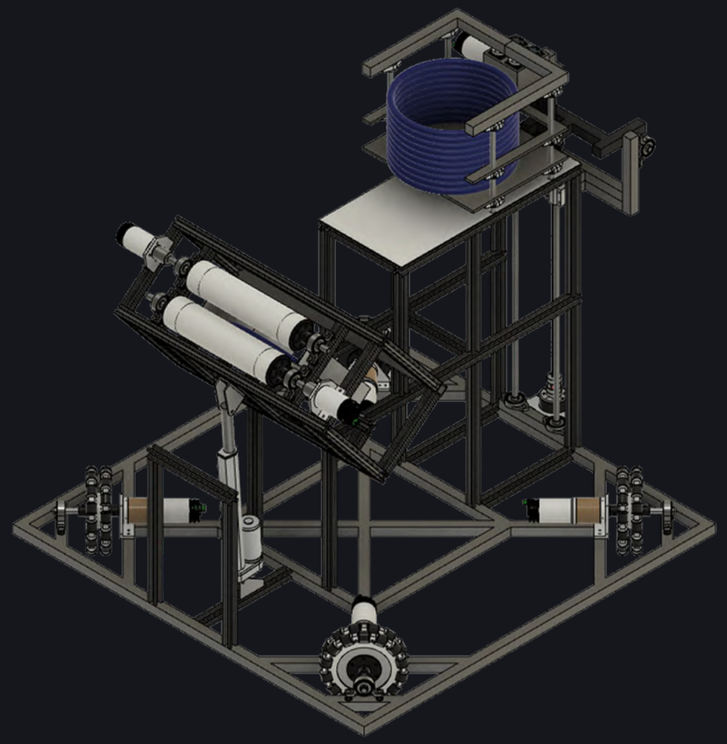
SELECTED PROJECTS

# ABU ROBOCON 2023 ROBOT





## DESCRIPTION:

**A Man-Operated Ring Throwing Robot**, capable of picking and throwing rings into poles.

## TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Grinding
- Drilling
- Hand Tools

## ROLE:

Mechanics (3D Design, Component Selection, Assembly)

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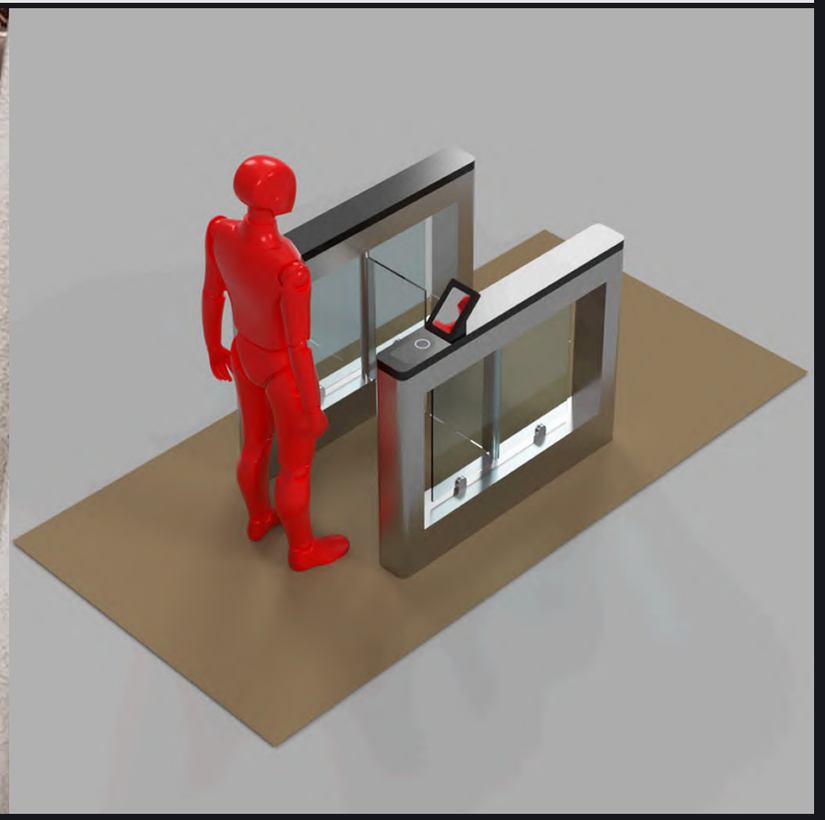
# SKYWARD PROJECT - ENTRANCE GATE

## DESCRIPTION:

**An Entrance Gate Concept and Mounted Display**, customized for ZKTeco Gate with a 3D printed display casing.

## TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Product Design
- Product Finishing
- Grinding
- Drilling
- Hand Tools



## ROLE:

3D Design Specialist - Mechatronics Engineer

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SELECTED PROJECTS

# SKYWARD PROJECT - KID'S GAME

## DESCRIPTION:

The Busy Table at Zeppelin, designed for kids' games from acrylics and 3D printed materials.

## TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Product Design
- Product Finishing
- Hand Tools

## ROLE:

3D Design Specialist - Mechatronics Engineer

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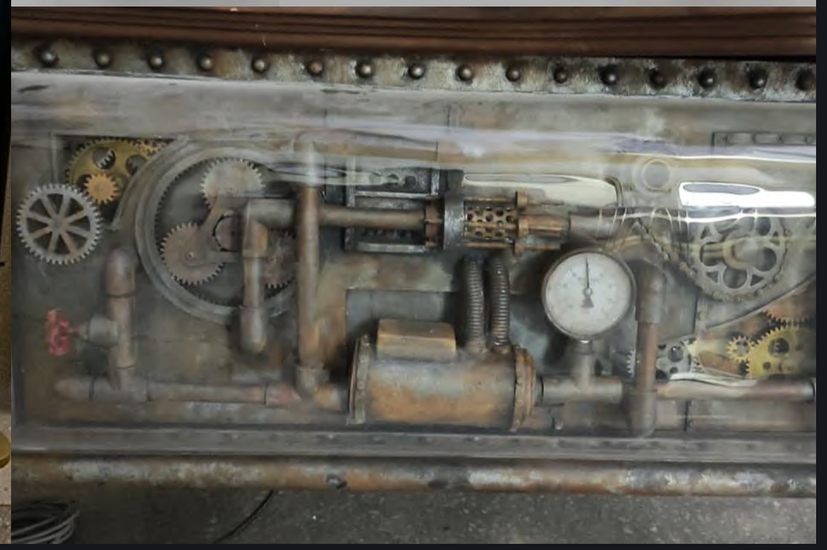
# SKYWARD PROJECT - CARD DISPENSER

## DESCRIPTION:

**A Card Dispensing Machine**, including gear sets as the machine's interactive display.

## TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Product Design
- Product Finishing



## ROLE:

3D Design Specialist - Mechatronics Engineer

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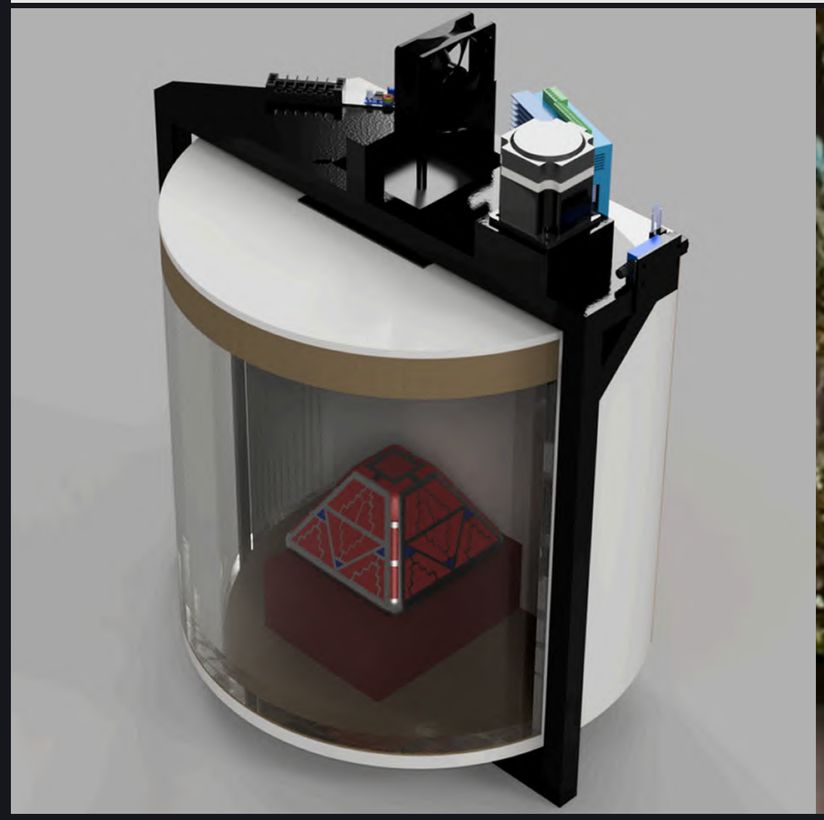
SELECTED PROJECTS

# MECHATRONICS DISPLAY AT MUNAS



## DESCRIPTION:

**An Automatic Artifact Display**, using six synchronized stepper motors for a Museum Nasional event.

## TOOLS & SKILLS:

- 3D CAD
- Mechanism Design
- Microcontroller
- C++ Programming

## ROLE:

Mechatronics Engineer (3D Design & Programming)

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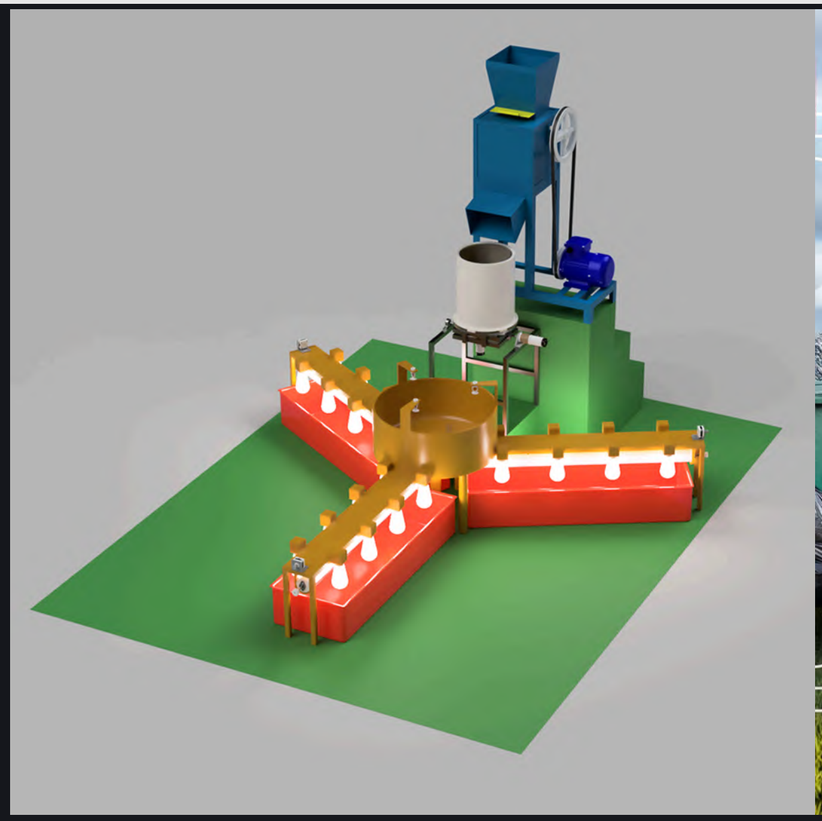
# MAGGOT FEEDER FOR INNOVILLAGE CLIENT

## DESCRIPTION:

**Maggofeed: A Maggot Feeder** for an organic waste processing system.

## TOOLS & SKILLS:

- 3D CAD
- Mechanism Design
- Product Design
- Detailed Blueprints



INNOVILLAGE

## ROLE:

Mechatronics Engineer (3D Design Concept)

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SELECTED PROJECTS

# AUTOMATIC CAT FEEDER PROTOTYPE

## DESCRIPTION:

**An Automatic Cat Feeder**, using sensors, a motor, and an ESP32 connected to Kodular and Antares for an IoT project.

## TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Product Design

## ROLE:

3D Printing Engineer (3D Design, 3D Printing, Assembly)

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Contacts:

See More:

SELECTED PROJECTS

# RADAR SLIDER FOR RESEARCH

## DESCRIPTION:

**An Automatic Mechatronics Slider**, a radar/reflector device supporting state-of-the-art radar research.

## TOOLS & SKILLS:

- 3D CAD
- Mechanism Design
- Grinding
- Drilling
- Hand Tools

## ROLE:

Mechatronics Engineer (3D Design & Assembly)

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# SELECTED PROJECTS

## 1. ABU ROBOCON ROBOTS

- Fully Self-Designed and Built Robots
- Competed at the National Stage of Indonesian Robocon (KRI)
- Full Game Points in Regional Qualification
- 4th Place Finish in the 2024 National Stage

Link: abu-robocon-2024
abu-robocon-2023

## 2. SKYWARD PROJECT

- 3D Design Specialist at Arthatronic Studio
- Designed and Manufactured for 3D Printing and Acrylics
- Executed Client-Specified Requests
- Contributed to Project Phases 1 & 2

Link: skyward-project

## 3. MECHATRONICS DISPLAY

- Full Component Selection and Integration
- Designed with Blueprints for Manufacturing
- Programmed and Synchronized Six Motors Seamlessly
- Featured at Munas Repopening Event

Link: mechatronics-artifact-display

## 4. MAGGOT FEEDER

- Client Project for Innovillage Team
- Full Component Selection for Blueprints
- Developed Design Concept with Blueprints for Manufacturing
- Recognized by Telkom University

Link: maggot-feeder

## 5. AUTOMATIC CAT FEEDER

- Fully Self-Designed and Built Mechanism
- Incorporated Sensors, Actuators, and Microcontroller Integration
- Optimized Design for 3D Printing
- Achieved A Grade in IoT Class

Link: automatic-cat-feeder

## 6. RADAR SLIDER

- Supported State-of-The-Art Radar Research
- Fully Self-Designed and Built Mechanism
- Incorporated Sensors, Actuators, and Microcontroller Integration
- Adaptive Design for Radar Size Adjustment

Link: radar-slider

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# OTHER PROJECTS

Link: other-projects

## SMART WATCH

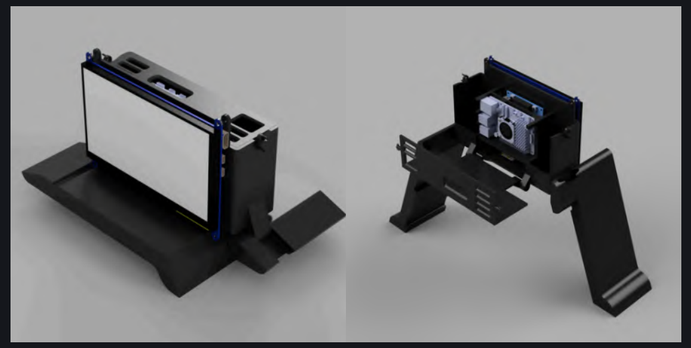
An arrhythmia detection watch.

## PANEL BOX

Custom electronic panel boxes with displays.

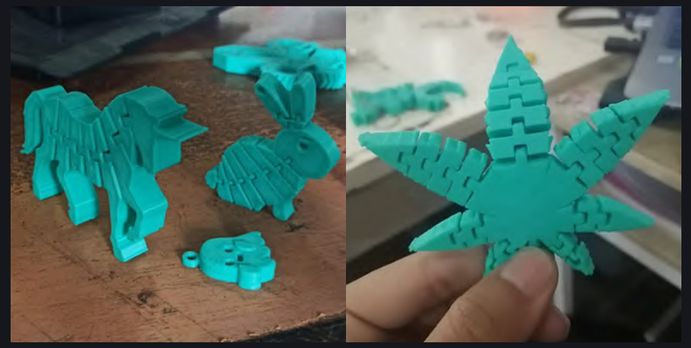
## GRIM REAPER

A 3D printed grim reaper figurine.



## PROTABLE CAMERA DEVICE

A vena detection device using raspi camera.



## FLEXIBLE KEYCHAINS

Various 3D printed flexible keychains.

## LINE FOLLOWER ROBOT

An educational modular robot concept.

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# OTHER PROJECTS

Link: other-projects

**ACCESSORIES, MINIATURES, TOYS, KEYCHAINS, MOCKUPS, PROTOTYPES, ETC**

*and more design & printing requests.....*

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Contacts:

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# CONTACTS & LINKS

More documentations, details, price lists, and free consultations, reach us on:

**WhatsApp:** wa.me/+6282182227816

**Instagram:** instagram.com/ariqshaq

**LinkedIn:** linkedin.com/in/ariqshaq

**GitHub:** github.com/ariqshaq/3d-design-portfolio

"DESIGNING SOLUTIONS FOR YOUR VISION"

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3D DESIGN & 3D PRINTING SOLUTIONS